

Learn by Doing: Number Sense

UTSA Problem Solving Club • Fall 2026

Unsimplified answers don't count for much!

Each of these has a specific numerical answer, and none of them needs a calculator.

- (a) Simplify $\sqrt{7 + 4\sqrt{3}}$.
(b) Real numbers x and y satisfy $x + y = 5$ and $x^3 + y^3 = 35$. Find xy .
(c) Evaluate $\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{9}\right)\left(1 - \frac{1}{16}\right) \cdots \left(1 - \frac{1}{100}\right)$.

Powers of (Almost) Equal

Without a calculator, order the numbers 50^{50} , 49^{51} and 51^{49} from smallest to largest.

From hundred, to googol, to googolplex

(I) A *googol* is $G = 10^{100}$, i.e. a 1 followed by one hundred 0s. How many digits does each of these have?

- (a) $1 + 2 + \cdots + G$ (b) $1^2 + 2^2 + \cdots + G^2$ (c) $1^3 + 2^3 + \cdots + G^3$ (d) $1^{100} + 2^{100} + \cdots + G^{100}$
(e) $1^G + 2^G + \cdots + 100^G$ (f) $1^G + 2^G + \cdots + G^G$

(II) A *googolplex* is the number $\mathbb{G} = 10^G$ (a digit 1 followed by a googol zeros.) Redo part (I) with
(i) $\mathbb{G}$ in place of G everywhere, (ii) G in place of 100 everywhere, (iii) both $\mathbb{G}$ in place of G , and G in place of 100 everywhere.

Sum of digits of sum of digits of sum of digits of...

Let $N = 2026^{2026}$. Compute the sum of the digits of the sum of the digits of the sum of the digits of N .

Integer sequences in rationals!

- (a) Find the decimal expansion of $1/9899$. There is a clear pattern —explain *why* it arises.
(b) Find a fraction a/b , with a and b small, such that $a/b = 0.010203040506070809101112\dots$; then do the same for $0.010409162536\dots$.
(c) Compute the decimal expansion of $1/p$ for as many small primes $p = 7, 11, 13, 17, 19, \dots$ as you like. The expansion is always purely periodic; can you predict the *length* of the period before computing it? What changes in a base other than ten?

Upstream, Downstream

A boat trip down the Amazon from Santarém to Macapá takes 4 days going downstream, and the return trip upstream takes 6 days. Assume the current has the same speed all along the route, and that the boat's own speed through the water is the same on both legs.

How long does the trip take drifting downstream with no propulsion at all?

Competition-Level Problems

The Little Grasshopper that Could

A grasshopper starts at the origin in the coordinate plane and makes a sequence of hops. Each hop has length 5, and after each hop the grasshopper is at a point whose coordinates are both integers; thus, there are 12 possible locations for the grasshopper after the first hop. What is the smallest number of hops needed for the grasshopper to reach the point $(2021, 2021)$?

Source: Putnam 2021 A1

Every Composite, Almost a Product

Show that every composite positive integer is expressible as $xy + yz + zx + 1$, with x, y, z positive integers.

Source: Putnam 1988 B1

Two, Three, Four

Determine all positive integers n for which there exist positive integers a, b , and c satisfying $2a^n + 3b^n = 4c^n$.

Source: Putnam 2024 A1

d-d-death (to 2016)

Find the smallest positive integer j such that for every polynomial $p(x)$ with integer coefficients and for every integer k , the integer

$$p^{(j)}(k) \quad (\text{the } j\text{-th derivative of } p(x) \text{ at } k)$$

is divisible by 2016.

Source: Putnam 2016 A1

$A^3 + B^3 + C^3 - 3ABC = ?$

Determine all possible values of the expression

$$A^3 + B^3 + C^3 - 3ABC$$

where A, B , and C are nonnegative integers.

Source: Putnam 2019 A1

What's up, $\mathbb{F}_{1979}$?

Write

$$\frac{p}{q} = 1 - \frac{1}{2} + \frac{1}{3} - \cdots + \frac{1}{1319}$$

as a reduced fraction with integers $p, q > 0$. Prove that the numerator p is divisible by 1979.

Source: IMO 1979, Problem 1